

Machine Learning

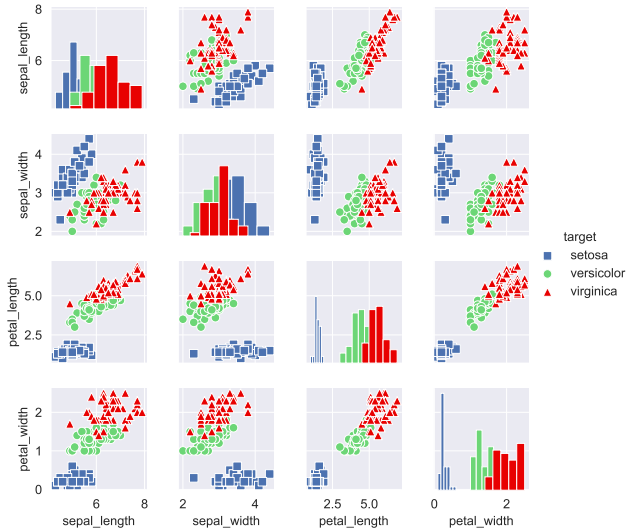
Lecture 2: k -Nearest Neighbors

Dr. Leo Schwinn

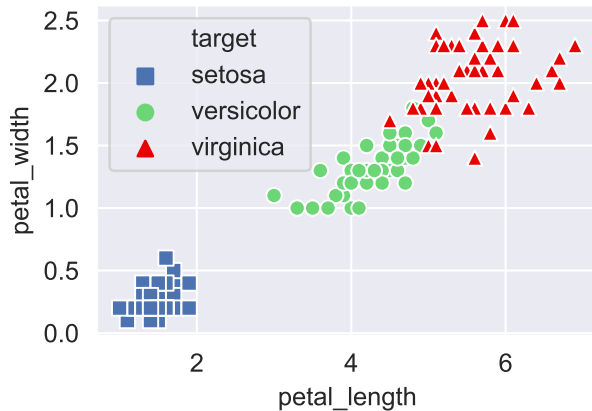
Data Analytics and Machine Learning
Technical University of Munich

Winter term 2024/2025

Iris dataset



Iris dataset: 2 features



How do we intuitively label new samples by hand?
Look at the *surrounding* points. Do as your **neighbor** does.

1-NN algorithm

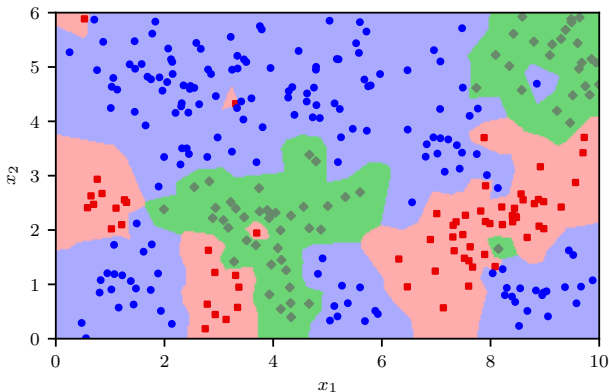
Given a training dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$
where $\mathbf{x}_i \in \mathbb{R}^D$ are features and $y_i \in \{1, \dots, C\}$ are class labels

To classify new observations:

- define a distance measure (e.g. Euclidean distance)
- compute the nearest neighbor for all new data points
- and label them with the label of their nearest neighbor

This works for both *classification* and *regression*.

1-NN



This corresponds to a Voronoi tessellation.
And results in poor generalization...

k -Nearest Neighbor classification

More *robust* against errors in the training set:

Look at multiple nearest neighbors and pick the **majority** label.

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Let $\mathcal{N}_k(\mathbf{x})$ be the k nearest neighbors of a vector $\mathbf{x}$, then in classification tasks:

$$p(y = c \mid \mathbf{x}, k) = \frac{1}{k} \sum_{i \in \mathcal{N}_k(\mathbf{x})} \mathbb{I}(y_i = c),$$

$$\hat{y} = \arg \max_c p(y = c \mid \mathbf{x}, k)$$

with the *indicator variable* $\mathbb{I}(e)$ is defined as:

$$\mathbb{I}(e) = \begin{cases} 1 & \text{if } e \text{ is true} \\ 0 & \text{if } e \text{ is false.} \end{cases}$$

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i.e., the vector will be labeled by the mode of its neighbors' labels.

Refresher: discrete probability theory

Given a jar that contains different colored balls $\{4, 10, 6\}$. What is the probability of randomly drawing a ball with a particular color (e.g. red)?

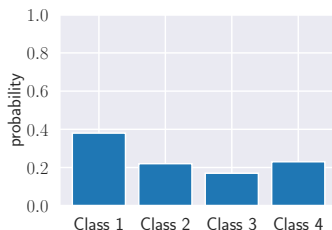
Intuitively: $p(\text{ball} = \text{red}) = \frac{\text{number of red balls}}{\text{total number of balls}} = \frac{4}{4+10+6} = \frac{4}{20} = 0.2$

Similarly: $p(\text{ball} = \text{green}) = 0.5$, $p(\text{ball} = \text{blue}) = 0.3$

The probability mass function p assigns value to each possible outcome.

In general it has to hold:

- $\forall x, p(X = x) \geq 0$
- $\sum_x p(X = x) = 1$



k -Nearest Neighbor classification: weighted

Look at multiple nearest neighbors and pick the **weighted majority** label.

k -Nearest Neighbor classification: weighted

Look at multiple nearest neighbors and pick the **weighted majority** label. The weight is **inversely proportional** to the distance.

Let $\mathcal{N}_k(\mathbf{x})$ be the k nearest neighbors of a vector $\mathbf{x}$, then in classification tasks:

$$p(y = c \mid \mathbf{x}, k) = \frac{1}{Z} \sum_{i \in \mathcal{N}_k(\mathbf{x})} \frac{1}{d(\mathbf{x}, \mathbf{x}_i)} \mathbb{I}(y_i = c),$$

$$\hat{y} = \arg \max_c p(y = c \mid \mathbf{x}, k)$$

with $Z = \sum_{i \in \mathcal{N}_k(\mathbf{x})} \frac{1}{d(\mathbf{x}, \mathbf{x}_i)}$ the normalization constant and $d(\mathbf{x}, \mathbf{x}_i)$ being a distance measure between $\mathbf{x}$ and $\mathbf{x}_i$.

k -Nearest-Neighbor regression

Regression is similar:

Let $\mathcal{N}_k(\mathbf{x})$ be the k nearest neighbors of a vector $\mathbf{x}$, then for regression:

$$\hat{y} = \frac{1}{Z} \sum_{i \in \mathcal{N}_k(\mathbf{x})} \frac{1}{d(\mathbf{x}, \mathbf{x}_i)} y_i,$$

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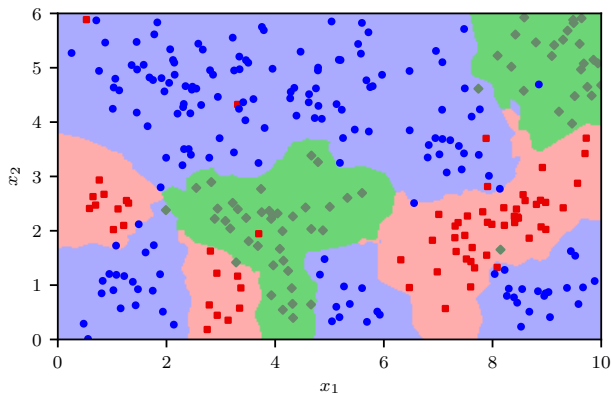
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i.e., the vector will be labeled by a **weighted mean** of its neighbors' values.

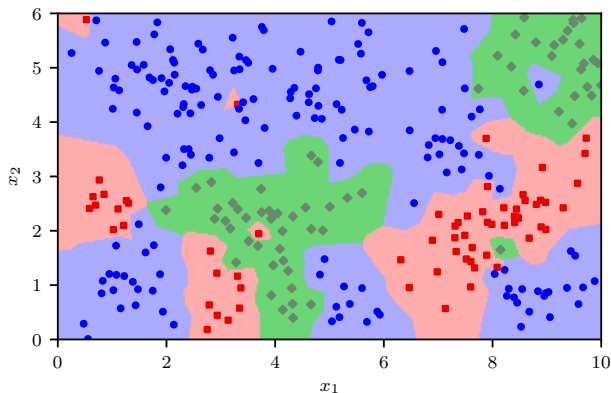
Note: y_i is a real number here (rather than categorical label).

3-NN



So, how many neighbors are best?

1-NN

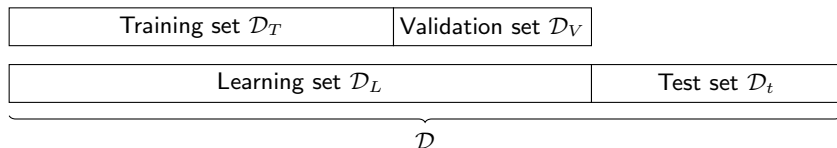


Compare the decision boundaries of 1-NN and 3-NN

Choosing k

Goal is **generalization**: pick k (called a *hyper-parameter*) that performs best¹ on unseen (future) data.

Unfortunately, no access to future data, so split the dataset $\mathcal{D}$:



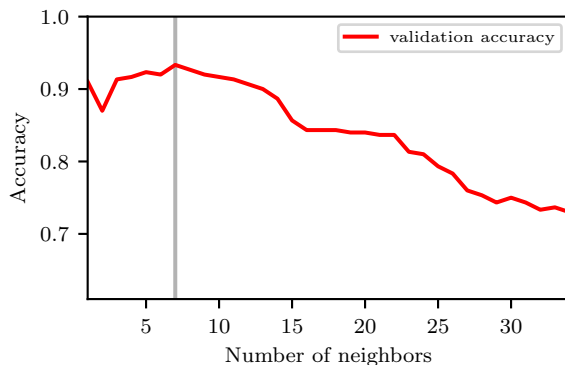
Hyper-parameter tuning procedure

- Learn the model using the training set
- Evaluate performance with different k on the *validation set* picking the best k
- Report final performance on the test set.²

¹In terms of some predefined metric, e.g., accuracy

²Good data science practices: See slides on Decision Trees

Using validation set to choose k



We choose $k = 7$.

Measuring classification performance

How can we assess the performance of a (binary) classification algorithm?

⇒ Confusion table

Predicted	True condition	
	$y = 1$	$y = 0$
$y = 1$	TP	FP
$y = 0$	FN	TN

TP = true positive
 TN = true negative
 FP = false positive
 FN = false negative

} correct predictions

} wrong predictions

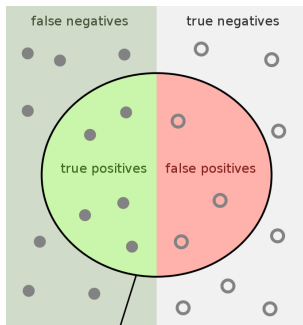


Image source: https://en.wikipedia.org/wiki/Precision_and_recall

Measuring classification performance

Accuracy:
$$\text{acc} = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision:
$$\text{prec} = \frac{TP}{TP + FP}$$

Sensitivity/Recall:
$$\text{rec} = \frac{TP}{TP + FN}$$

Specificity:
$$\text{tnr} = \frac{TN}{FP + TN}$$

False Negative Rate:
$$\text{fnr} = \frac{FN}{TP + FN}$$

False Positive Rate:
$$\text{fpr} = \frac{FP}{FP + TN}$$

F1 Score:
$$f1 = \frac{2 \cdot \text{prec} \cdot \text{rec}}{\text{prec} + \text{rec}}$$

⇒ Trade-off between precision and recall: increasing one (most often) leads to decreasing the other

General note: Be careful when you have imbalanced classes!

Distance measures

- K-NN can be used with various distance measures $\rightarrow$ highly flexible
- Euclidean distance (L_2 norm): $\sqrt{\sum_i (u_i - v_i)^2}$

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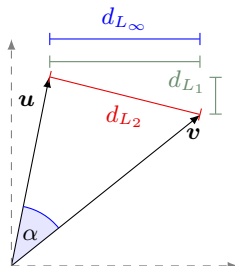
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- Angle:

$$\cos \alpha = \frac{\mathbf{u}^T \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}$$

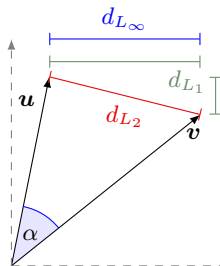


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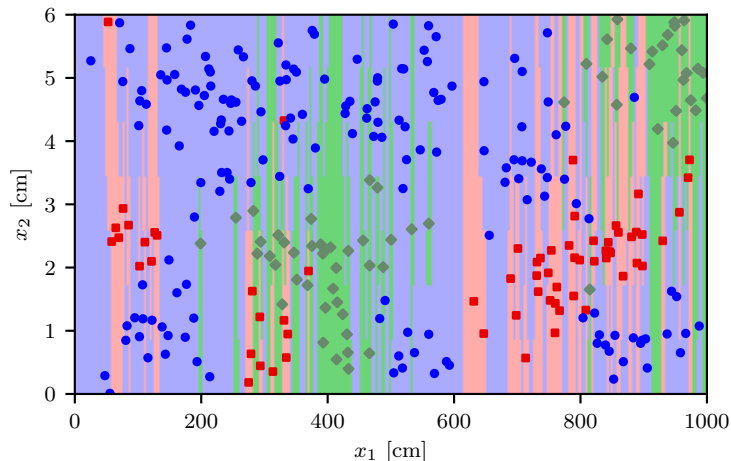


- Mahalanobis distance (Σ is positive (semi) definite and symmetric):

$$\sqrt{(\mathbf{u} - \mathbf{v})^T \Sigma^{-1} (\mathbf{u} - \mathbf{v})}$$

- Hamming distance, Edit distance, ...

Scaling issues



The same old example but one of our features is in the order of meters, the other in the order of centimeters. ($k = 1$)

Circumventing scaling issues

- Data *standardization*

Scale each feature to zero mean and unit variance.

$$x_{i,\text{std}} = \frac{x_i - \mu_i}{\sigma_i}$$

(This is a standard procedure in machine learning. Many models are sensitive to differences in scale.)

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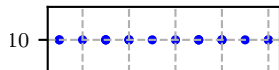
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- Use the Mahalanobis distance.

$$\text{mahalanobis}(\mathbf{x}_1, \mathbf{x}_2) = \sqrt{(\mathbf{x}_1 - \mathbf{x}_2)^T \mathbf{\Sigma}^{-1} (\mathbf{x}_1 - \mathbf{x}_2)}$$

$$\mathbf{\Sigma} = \begin{bmatrix} \sigma_1^2 & 0 & 0 \\ 0 & \dots & 0 \\ 0 & 0 & \sigma_n^2 \end{bmatrix} \text{ is equal to Euclidean distance on normalized data}$$

The curse of dimensionality



Given a discrete one-dimensional input space
 $x \in \{1, 2, \dots, 10\}$

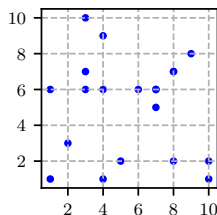
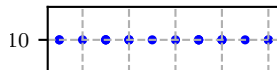
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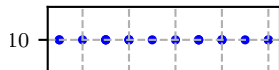
Add a second dimension (now
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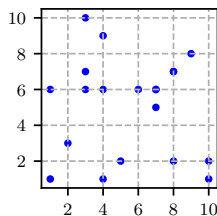
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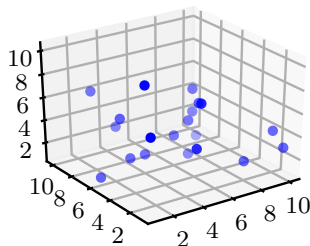
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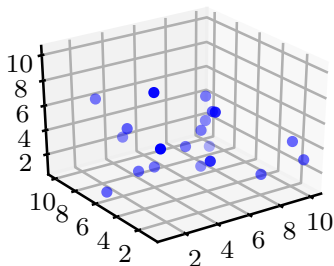


Once you add a third dimension you only cover 2%.



The curse of dimensionality

- The nearest neighbor will now be pretty far away..
- N has to grow exponentially with the number of features. Consider this when using k -NN on high-dimensional data.

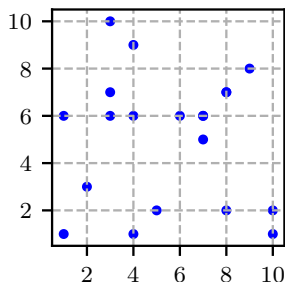


Practical considerations

Expensive: memory and naive inference are both $O(N)$:

we need to store the entire training data and compare with all training instances to find the nearest neighbor

Solution: use tree-based search structures (e.g. k-d tree) for efficient (approximate) NN³



³At the expense of an additional computation performed only once

What we learned

- k -NN Algorithm
- Train-validation-test split
- Measuring classification performance
- Distance metrics
- Curse of dimensionality

Reading material

Main reading

- "Machine Learning: A Probabilistic Perspective" by Murphy
[ch. 1.4.1 - 1.4.3]

Extra reading

- "Bayesian Reasoning and Machine Learning" by Barber
[ch. 14]

Slides adapted from previous versions by W. Koepp & D. Korhammer